

1, Explain behaviour of hub device

- * Hub is used to connect multiple network device
- * It is a Layer 1 (physical Layer) device.
- * It is a broadcast device
- * When data is received the hub sends it to all connected device
- * It does not learn or store MAC addresses
- * Because it broadcasts to all devices, Network traffic is high

2, Explain behaviour of switch device

- * Switch is used to connect multiple network device
- * It maintains a MAC address table.
- * It learns the MAC address and corresponding port
- * It is a Layer 2 (data Link Layer) device
- * If the destination MAC is known: It performs unicast Communication
- * If the destination MAC is unknown it performs broadcast
- * Switch reduces unnecessary traffic compared with a hub
- * Each switch port provides a separate collision domain
- * It is usually plug and play.

3, Explain behaviour of router device | but configuration can be done when req

- * Router is used to connect different network.
- * It is a Layer 3 (Network layer) device.
- * It uses IP address for forwarding packets
- * It maintains a routing table
- * The routing table contains information about available routes
- * Router forwards packets from one network to another network
- * It separates broadcast domain
- * Router mainly performs unicast communication
- * Routers generally require configuration for proper network communication.